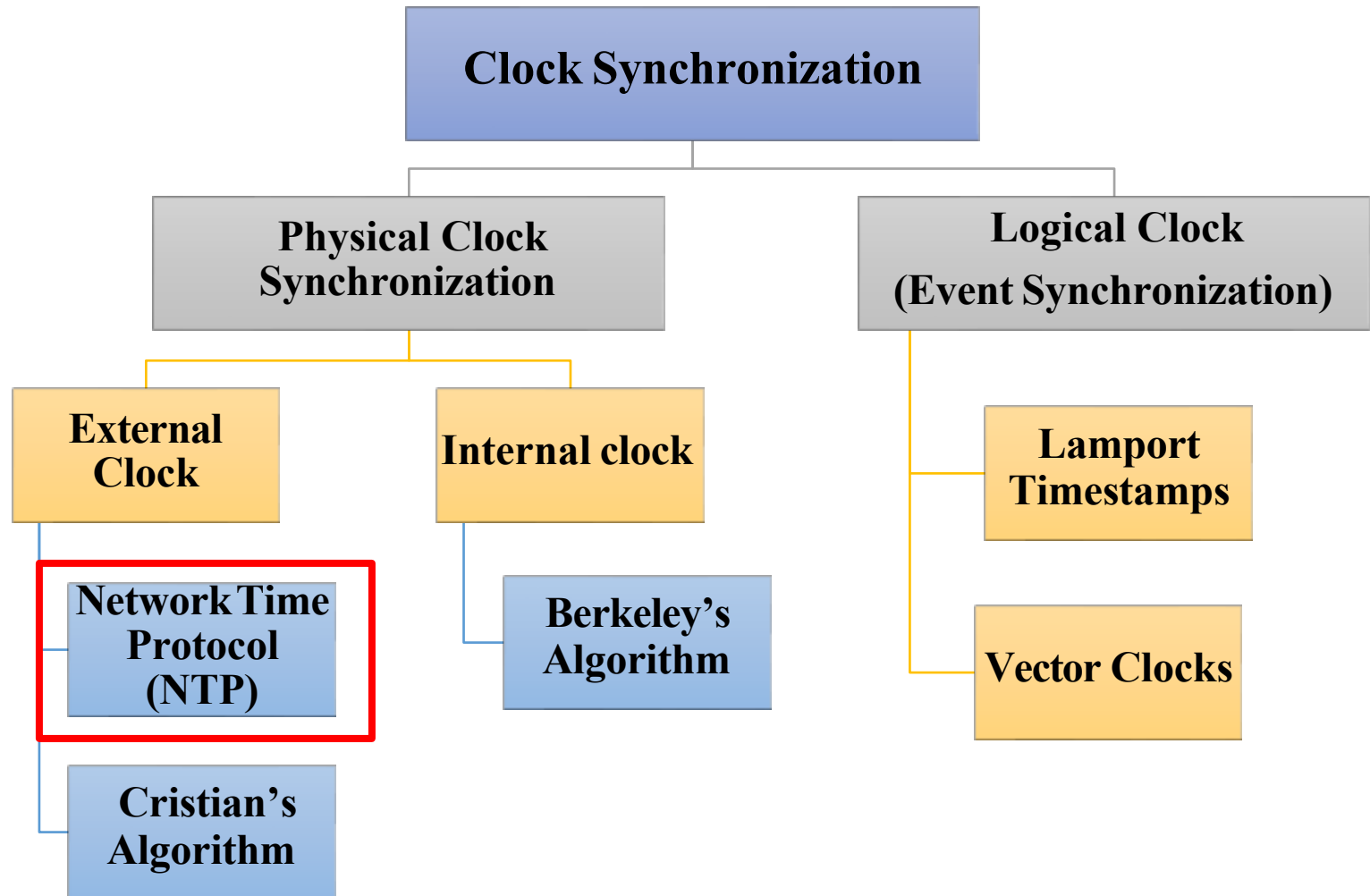
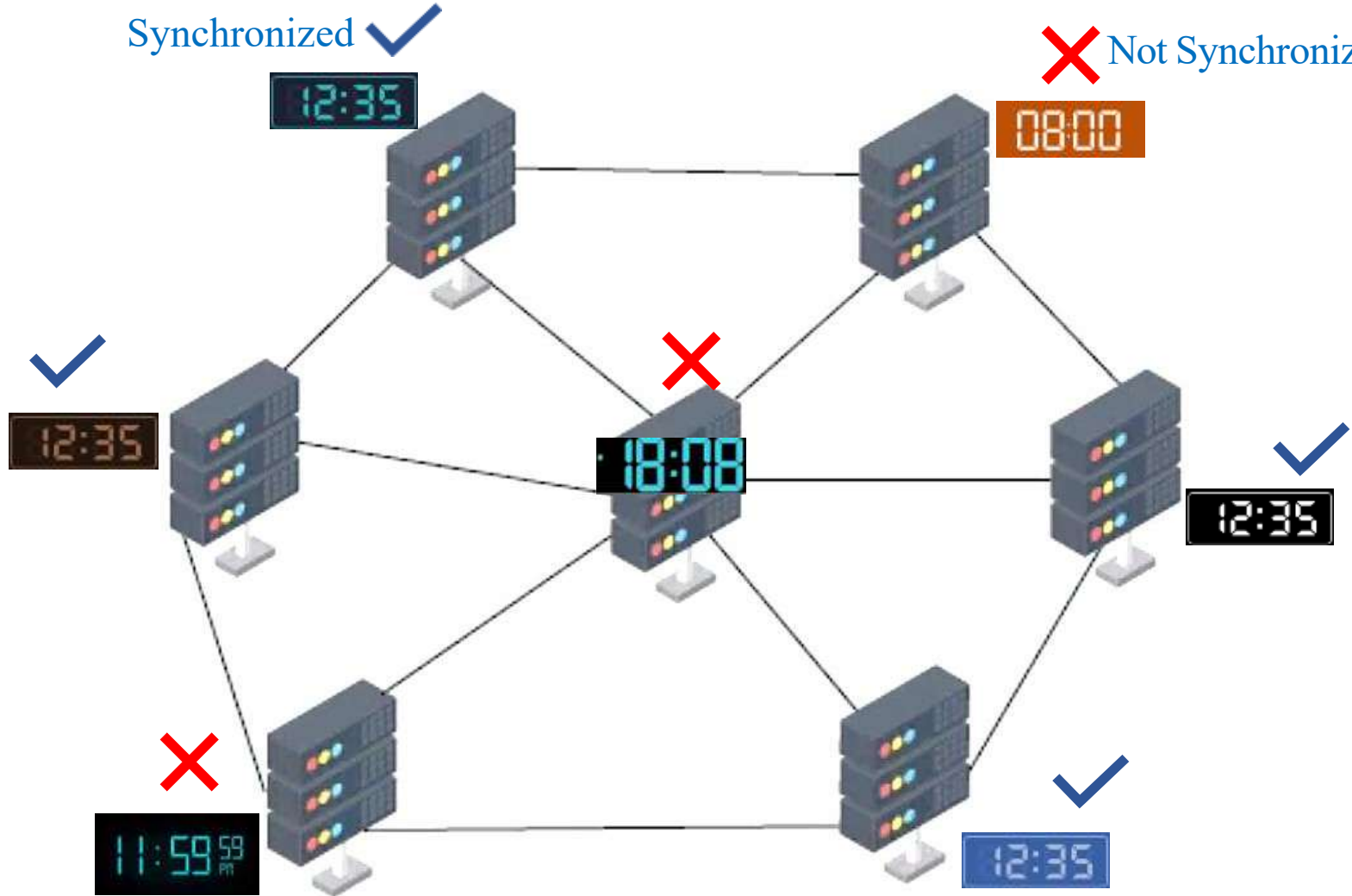




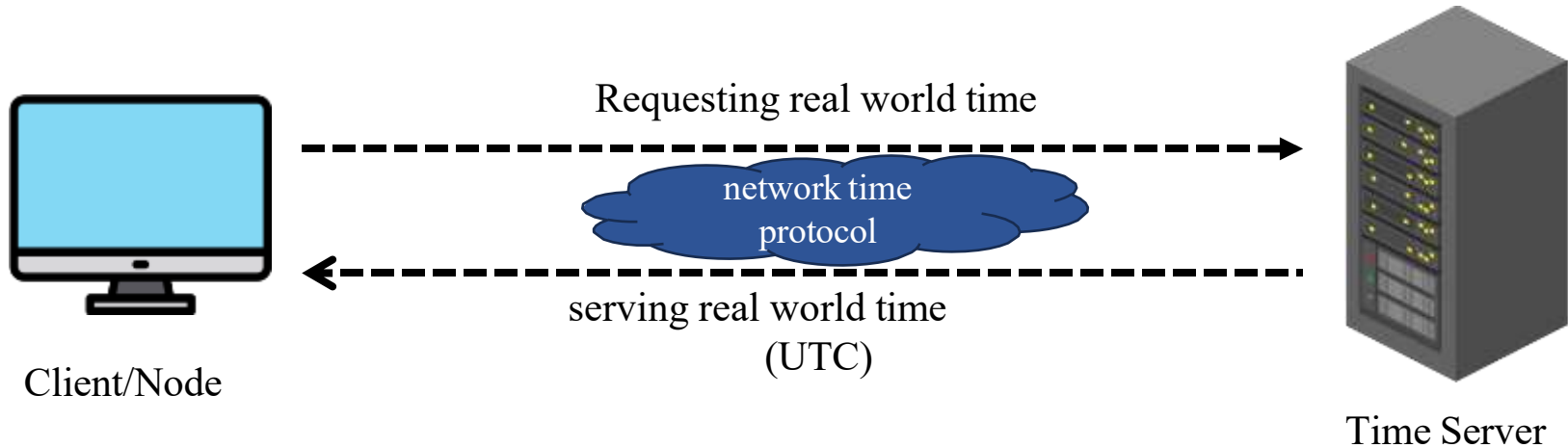
Synchronized ✓

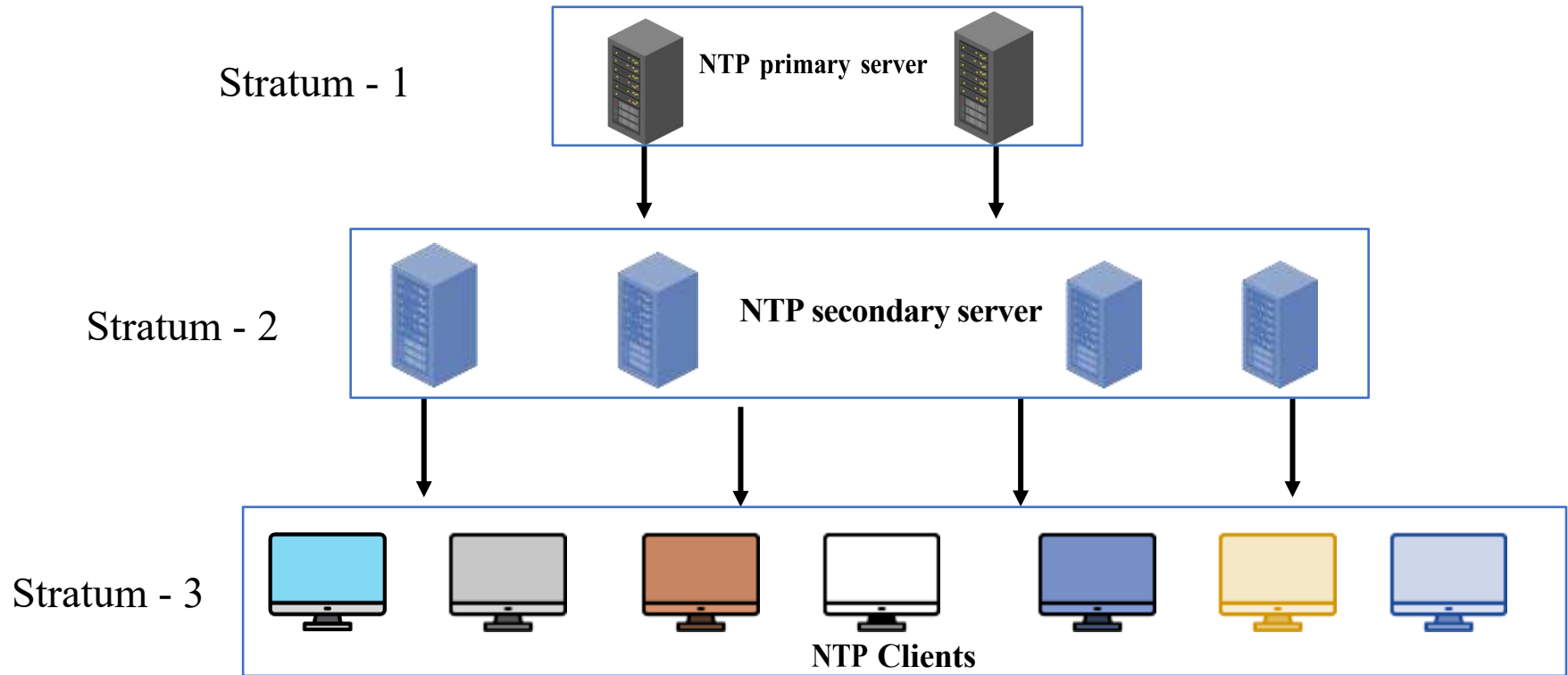
✗ Not Synchronized



Network Time Protocol

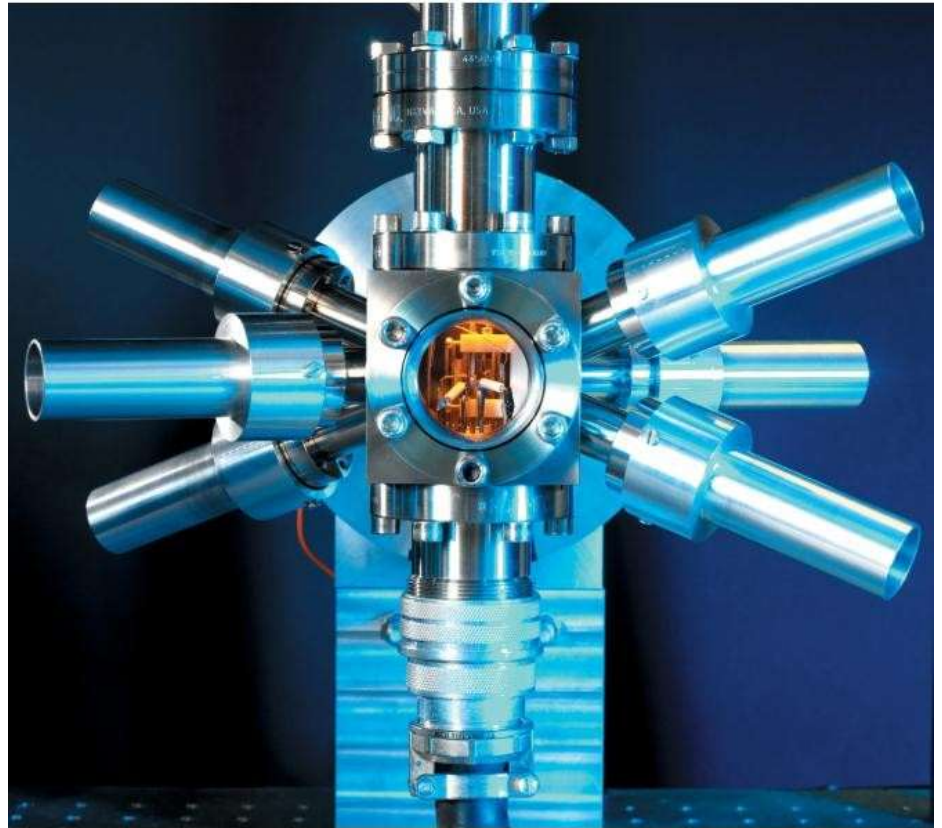
Network Time Protocol (NTP) is a protocol used to synchronize the clocks of computers over a network to a reference time such as **UTC (Coordinated Universal Time)**.







Caesium Atom

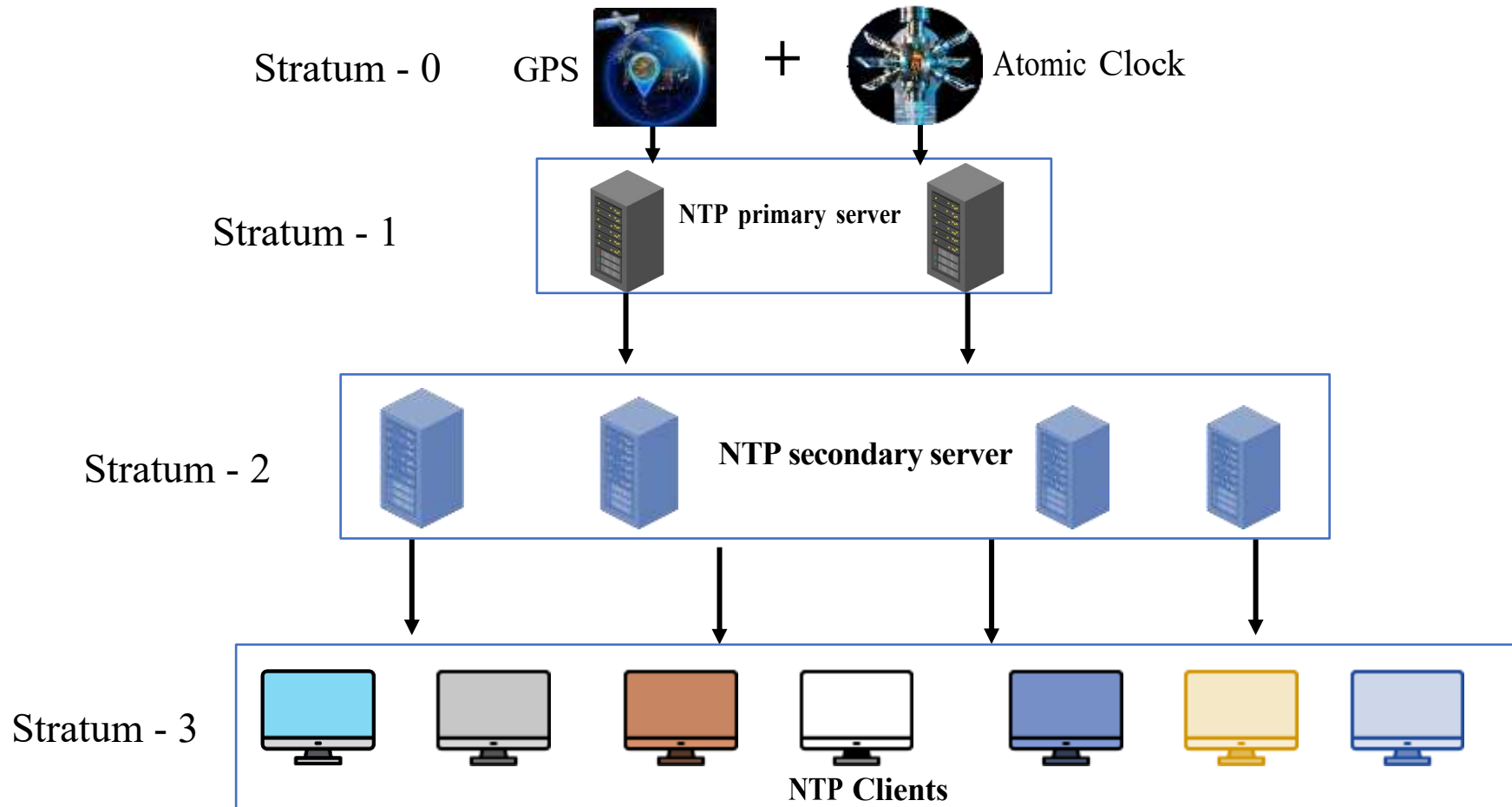


Atomic Clock

1 second = 9,192,631,770 vibrations of cesium

GPS Satellite





Stratum 0 – Reference Clocks (GPS + Atomic Clock)

- These are **highly accurate physical time sources**, such as **GPS receivers, atomic clocks, or radio clocks**.
- They generate the **original UTC time** that serves as the global reference.

Stratum 1 – Primary NTP Servers

- These are **primary time servers** directly connected to Stratum 0 reference clocks.
- They receive precise time and serve as **authoritative sources** for other servers.
- They use the **NTP protocol** to share accurate time with lower levels (Stratum 2).

Stratum 2 – Secondary NTP Servers

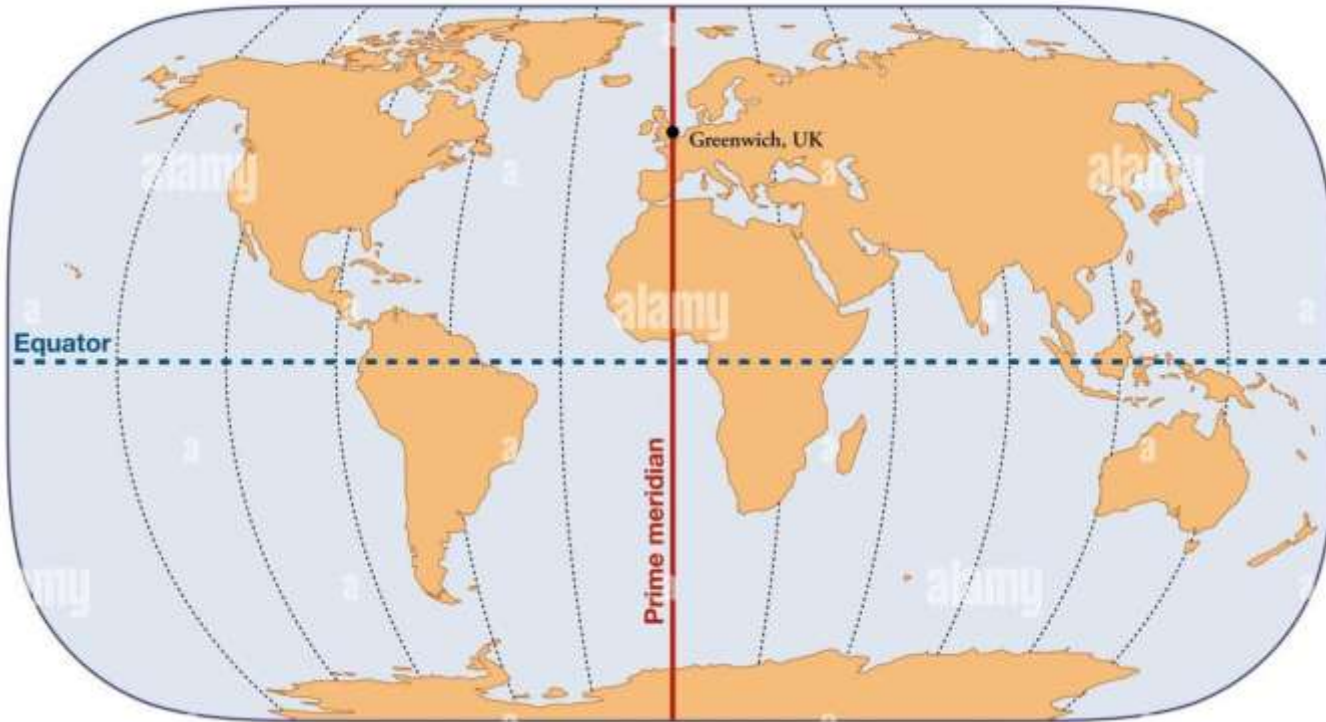
- These servers **synchronize their time** with one or more Stratum 1 servers.
- They act as **distributors** of time to large networks, organizations, or ISPs.
- Their main purpose is to **reduce the load** on primary servers and provide **redundancy**.

Stratum 3 – NTP Clients (End Systems)

- These are **ordinary computers, routers, or IoT devices** that get time from Stratum 2 servers.
- Clients **adjust their local clocks** based on the received UTC time.

What is UTC (Coordinated Universal Time)?

UTC is the **official global time standard** used by the entire world to keep clocks synchronized. Every country's time zone (like **IST = UTC +5:30**) is defined **relative to UTC**.



Time & language > Date & time

13:38

11 November 2025



Time zone

(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi



Region

India



Set time zone automatically

Off



Time zone

(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi



Set time automatically

On



Show time and date in the System tray

Turn this off to hide your time and date information on the taskbar

On



Change the date and time format



Show time in Notification Center

Off



